

EXP 12: Face Detection with Haar Cascades

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```
In [1]: import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: image = cv2.imread("saadhu.jpg")
if image is None:
    print("Image not found")
    exit()
# Extract ROI
roi = image[50:250, 50:250]
# Convert BGR to RGB
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
roi = cv2.cvtColor(roi, cv2.COLOR_BGR2RGB)
plt.subplot(1, 2, 1)
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1, 2, 2)
plt.imshow(roi)
plt.title("ROI")
plt.axis("off")
plt.show()
```

Original Image

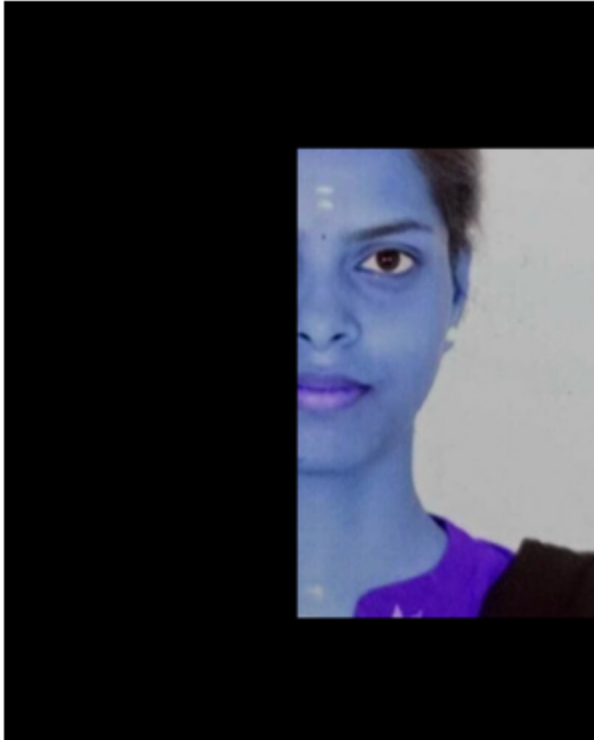


ROI



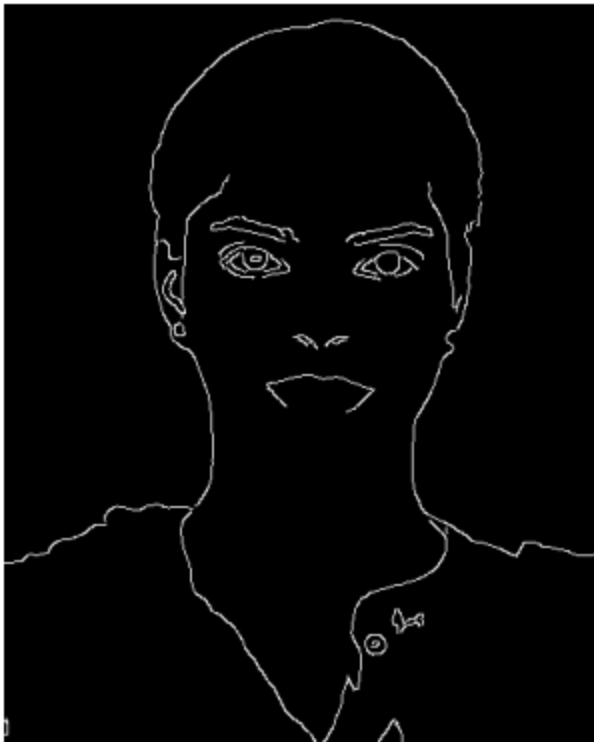
```
In [3]: roi = image[100:420, 200:550]
mask = np.zeros_like(image)
mask[100:420, 200:550] = roi
segmented = cv2.bitwise_and(image, mask)
plt.imshow(cv2.cvtColor(segmented, cv2.COLOR_BGR2RGB))
plt.title("Segmented ROI")
plt.axis('off')
plt.show()
```

Segmented ROI



```
In [4]: image = cv2.imread('saadhu.jpg')
if image is None:
    print("Error: dhoni.jpeg not found")
    exit()
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
edges = cv2.Canny(blur, 50, 150)
plt.imshow(edges, cmap='gray')
plt.title("Canny Edge Detection")
plt.axis('off')
plt.show()
```

Canny Edge Detection



```
In [5]: contours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
result = image.copy()
for c in contours:
    if cv2.contourArea(c) > 50:
        x, y, w, h = cv2.boundingRect(c)
        cv2.rectangle(result, (x, y), (x+w, y+h), (0, 255, 0), 2)
plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
plt.title("Contour Detection")
plt.axis('off')
plt.show()
```

Contour Detection

